

SP ENGINEERING ACADEMY

Video Lectures / Course Materials

COMPLETE COURSE

DIGITAL SYSTEMS

DIGITAL CIRCUIT DESIGN

Lecture Series #10 - Digital Converters

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Code Converters

I. 4-bit Binary to BCD Converter

Video Link - <https://youtu.be/499BK8JRGrA>

Input- Code : Binary Code : $B_3 B_2 B_1 B_0$ (LSB)

Output- Code : BCD (Decimal) Code : $D_4 D_3 D_2 D_1 D_0$ (LSB)

The truth table relating binary and BCD code

Binary Code $B_3 B_2 B_1 B_0$	Decimal Number	BCD Code $D_4 D_3 D_2 D_1 D_0$
0 0 0 0	0	0 0 0 0 0
0 0 0 1	1	0 0 0 0 1
0 0 1 0	2	0 0 0 1 0
0 0 1 1	3	0 0 0 1 1
0 1 0 0	4	0 0 1 0 0
0 1 0 1	5	0 0 1 0 1
0 1 1 0	6	0 0 1 1 0
0 1 1 1	7	0 0 1 1 1
1 0 0 0	8	0 1 0 0 0
1 0 0 1	9	0 1 0 0 1
1 0 1 0	10	1 0 0 0 0
1 0 1 1	11	1 0 0 0 1
1 1 0 0	12	1 0 0 1 0
1 1 0 1	13	1 0 0 1 1
1 1 1 0	14	1 0 1 0 0
1 1 1 1	15	1 0 1 0 1

For D_0

$B_3 B_2 \quad B_1 B_0$

	00	01	11	10
00			1	1
01			1	1
11			1	1
10			1	1

$$D_0 = B_0$$

For D_1

$B_3 B_2 \quad B_1 B_0$

	00	01	11	10
00			1	1
01			1	1
11	1	1		
10				

$$D_1 = B_3 B_2 \bar{B}_1 + \bar{B}_3 B_1$$

For D_2

$B_3 B_2 \quad B_1 B_0$

	00	01	11	10
00				
01	1	1	1	1
11			1	1
10				

$$D_2 = \bar{B}_3 B_2 + B_2 B_1$$

For D_3

$B_3 B_2 \quad B_1 B_0$

	00	01	11	10
00				
01				
11				
10	1	1		

$$D_3 = B_3 \bar{B}_2 \bar{B}_1$$

For D_4

$B_3 B_2 \quad B_1 B_0$

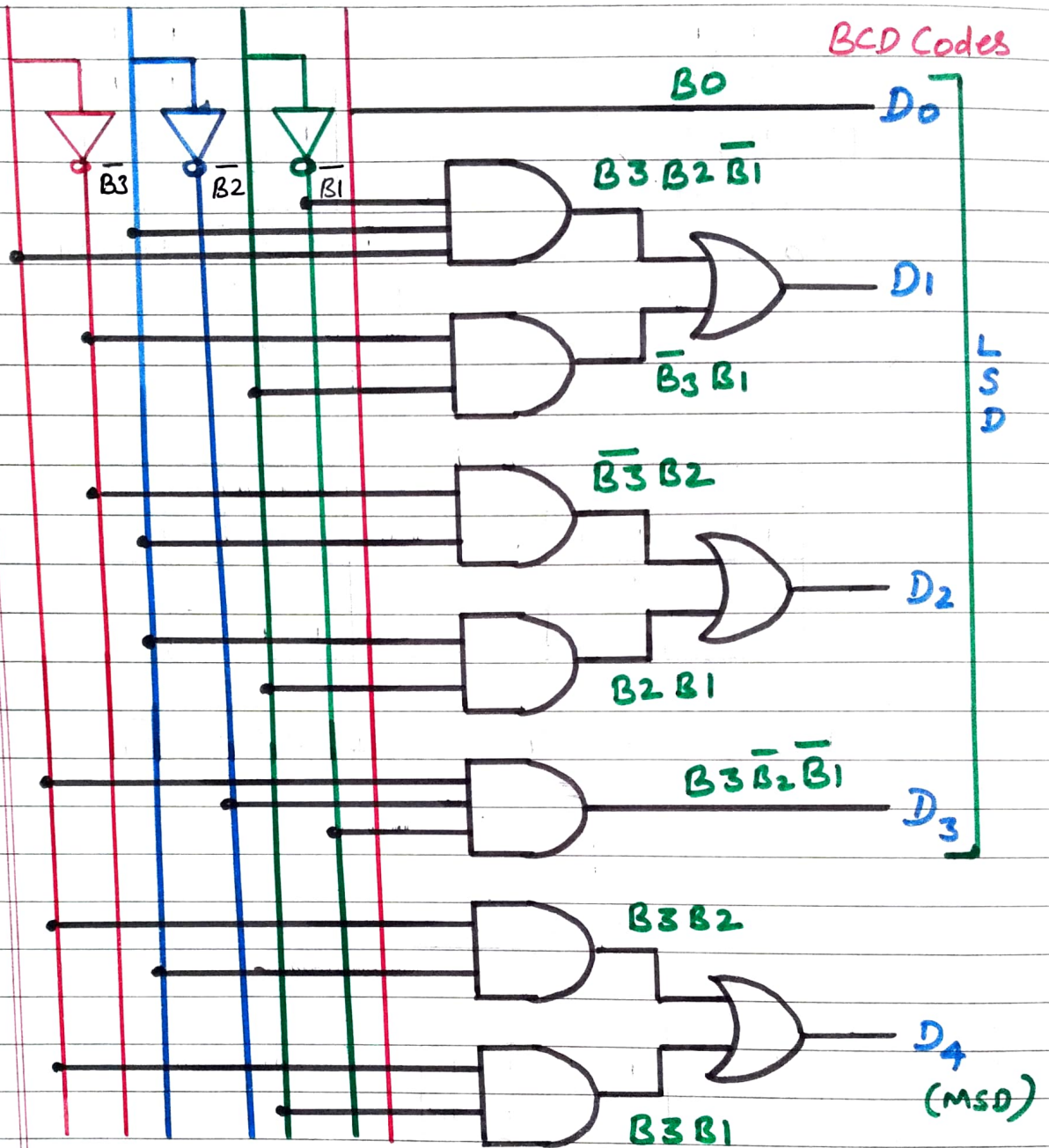
	00	01	11	10
00				
01				
11	1	1	1	1
10			1	1

$$D_4 = B_3 B_2 + B_3 B_1$$

Binary Codes

B3 B2 B1 B0

BCD Codes



II. 8421 BCD to Excess-3 Code Converter

Video Link - <https://youtu.be/oR0wtTxNy-8>

Excess-3 Code is a modified form of a BCD number. The excess-3 Code can be derived from the natural BCD Code by adding 3 to each coded number. For example, decimal 12 can be represented in BCD as 0001 0010. Now adding 3 to each digit we get Excess 3 Codes as 0100 0101 (45 in decimal).

Input Code : BCD Code : $D_3 D_2 D_1 D_0$ (D_0 LSB)

Output Code : Excess-3 Code : $E_3 E_2 E_1 E_0$ (E_0 LSB)

Decimal	BCD Code				Excess-3 Decimal	Excess-3 Code			
	D_3	D_2	D_1	D_0		E_3	E_2	E_1	E_0
0	0	0	0	0	0(3)	0	0	1	1
1	0	0	0	1	1(4)	0	1	0	0
2	0	0	1	0	2(5)	0	1	0	1
3	0	0	1	1	3(6)	0	1	1	0
4	0	1	0	0	4(7)	0	1	1	1
5	0	1	0	1	5(8)	1	0	0	0
6	0	1	1	0	6(9)	1	0	0	1
7	0	1	1	1	7(10)	1	0	1	0
8	1	0	0	0	8(11)	1	0	1	1
9	1	0	0	1	9(12)	1	1	0	0

For E3

$D_3 D_2 \quad D_1 D_0$

	00	01	11	10
00				
01		1	1	1
11	x	x	x	x
10	1	1	x	x

$$E_3 = D_3 + D_2(D_0 + D_1)$$

$$E_3 = D_3 + D_2 D_0 + D_2 D_1$$

For E2

$D_3 D_2 \quad D_1 D_0$

	00	01	11	10
00		1	1	1
01	1			
11	x	x	x	x
10		1	x	x

$$E_2 = D_2 \bar{D}_1 \bar{D}_0 + \bar{D}_2 (D_0 + D_1)$$

$$E_2 = D_2 \bar{D}_1 \bar{D}_0 + \bar{D}_2 D_0 + \bar{D}_2 D_1$$

For E1

$D_3 \quad D_2 \quad D_1 D_0$

	00	01	11	10
00	1		1	
01	1		1	
11	x	x	x	x
10	1		x	x

$$E_1 = \bar{D}_1 \bar{D}_0 + D_1 D_0$$

$$E_1 = D_1 \odot D_0$$

For E0

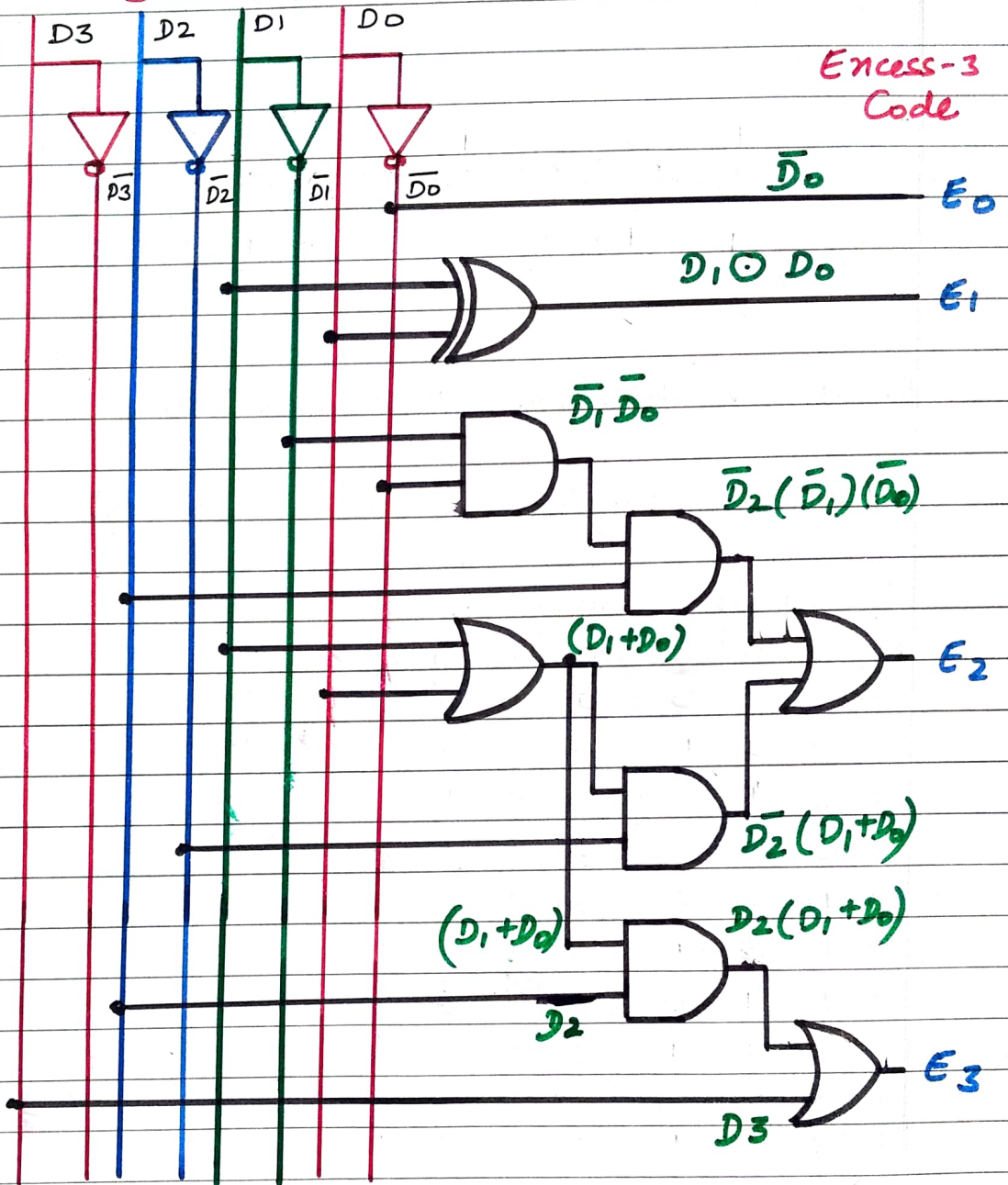
$D_3 \quad D_2 \quad D_1 D_0$

	00	01	11	10
00	1			1
01	1			1
11	x	x	x	x
10	1		x	x

$$E_0 = \bar{D}_0$$

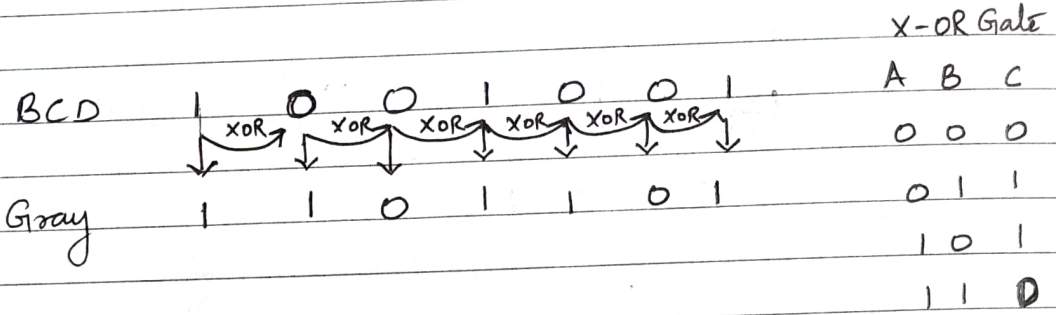
BCD Code

Excess-3 Code



III. BCD to Gray Code Converter

Video Link - https://youtu.be/d-qY_tdYxCw



Input Code : BCD Code : $D_3 D_2 D_1 D_0$ (D_0 LSB)

Output Code : Gray Code : $G_3 G_2 G_1 G_0$ (G_0 LSB)

BCD Code				Gray Code			
D_3	D_2	D_1	D_0	G_3	G_2	G_1	G_0
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1
0	0	1	0	0	0	1	1
0	0	1	1	0	0	1	0
0	1	0	0	0	1	1	0
0	1	0	1	0	1	1	1
0	1	1	0	0	1	0	1
0	1	1	1	0	1	0	0
1	0	0	0	1	1	0	0

For G_0

D_3	D_2	$D_0 D_1$				
		00	01	11	10	
00			1			1
01			1			1
11	x	x	x	x	x	x
10			1	x		x

For G_1

D_3	D_2	$D_1 D_0$			
		00	01	11	10
00				1	1
01			1	1	
11		x	x	x	x
10				x	x

$$G_0 = \bar{D}_1 D_0 + D_1 \bar{D}_0$$

$$G_0 = D_1 \oplus D_0$$

$$G_1 = D_2 \bar{D}_1 + \bar{D}_2 D_1$$

$$G_1 = D_2 \oplus D_1$$

For G_2

$D_3 \backslash D_2$	$D_1 D_0$	00	01	11	10
00					
01		1	1	1	1
11		x	x	x	x
10		1	1	x	x

$$G_2 = D_2 + D_3$$

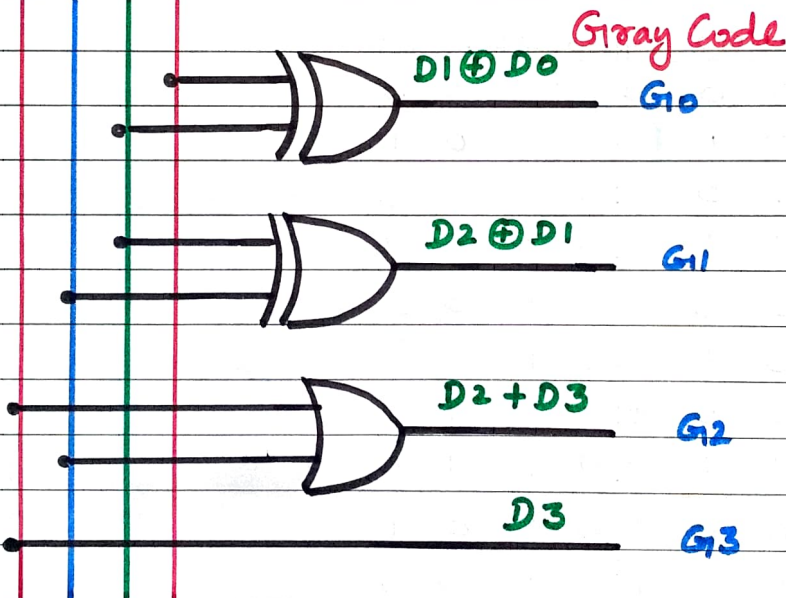
For G_3

$D_3 \backslash D_2$	$D_1 D_0$	00	01	11	10
00					
01					
11		x	x	x	x
10		1	1	x	x

$$G_3 = D_3$$

BCD Code

$D_3 \ D_2 \ D_1 \ D_0$



V. Excess-3 to BCD Code Converter

Video Link - <https://youtu.be/I9iEPBmYuP4>

Input Code: Excess-3 : $E_3 E_2 E_1 E_0$

Output Code: BCD : $D_3 D_2 D_1 D_0$

Decimal	Excess-3				Decimal	BCD			
	E_3	E_2	E_1	E_0		D_3	D_2	D_1	D_0
3	0	0	1	1	0	0	0	0	0
4	0	1	0	0	1	0	0	0	1
5	0	1	0	1	2	0	0	1	0
6	0	1	1	0	3	0	1	1	1
7	0	1	1	1	4	0	1	0	0
8	1	0	0	0	5	0	1	0	1
9	1	0	0	1	6	0	1	1	0
10	1	0	1	0	7	1	0	1	1
11	1	0	1	1	8	1	0	0	0
12	1	1	0	0	9	1	0	0	1

For D_3

$E_3 E_2$	$E_1 E_0$	00	01	11	10
00	X	X	0	X	
01	0	0	0	0	
11	1	X	X	X	
10	0	0	1	0	

For D_2

$E_3 E_2$	$E_1 E_0$	00	01	11	10
00	X	X	0	X	
01	0	0	1	0	
11	0	X	X	X	
10	1	1	0	1	

$$D_3 = E_3 E_2 + E_3 E_1 E_0$$

$$D_2 = E_2 E_1 E_0 + \bar{E}_2 \bar{E}_1 + \bar{E}_2 \bar{E}_0$$

For D_1

$E_3 E_2$	$E_1 E_0$	00	01	11	10
00	X	X	0	X	
01	0	1	0	1	
11	0	X	X	X	
10	0	1	0	1	

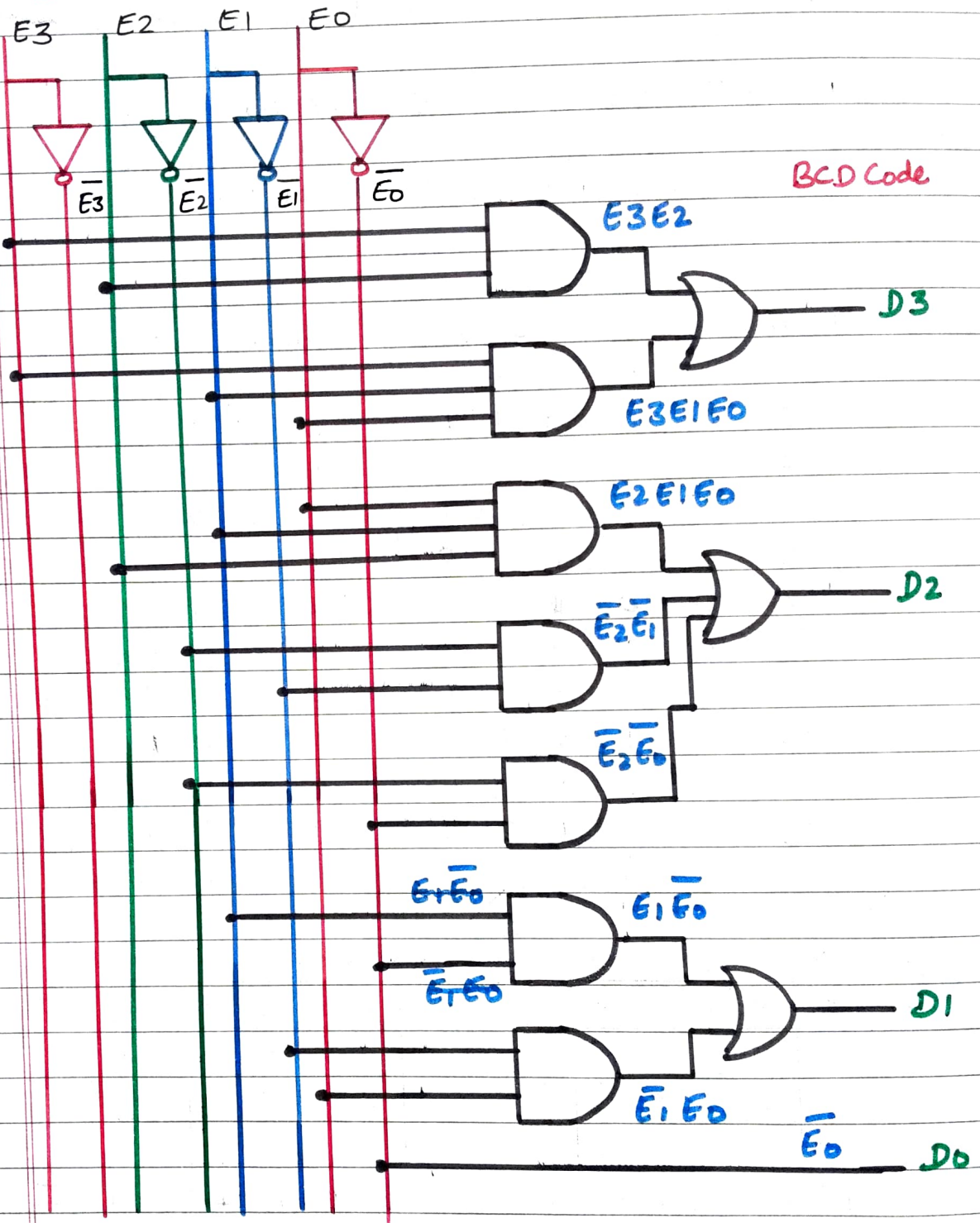
For D_0

$E_3 E_2$	$E_1 E_0$	00	01	11	10
00	X	X	0	X	
01	1	0	0	1	
11	1	X	X	X	
10	1	0	0	1	

$$D_1 = E_1 \bar{E}_0 + \bar{E}_1 E_0$$

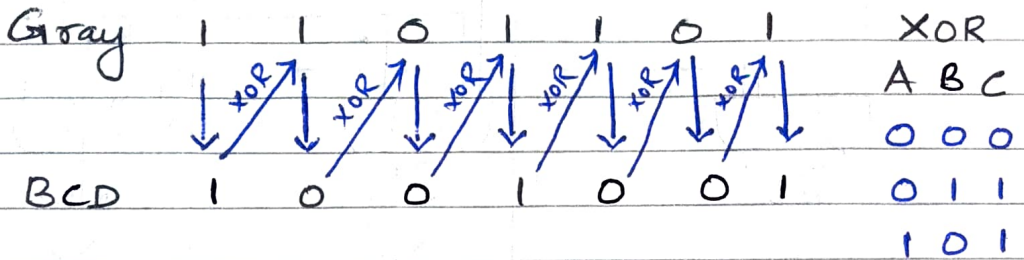
$$D_0 = \bar{E}_0$$

Excess-3 Code



Gray Code to BCD Converter

Video Link - <https://youtu.be/1kjLuEWck1U>



Input Code : Gray Code : $G_3 \ G_2 \ G_1 \ G_0$ (LSB)

Output Code : BCD : $B_3 \ B_2 \ B_1 \ B_0$ (LSB)

Gray Code				BCD			
G_3	G_2	G_1	G_0	B_3	B_2	B_1	B_0
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1
0	0	1	1	0	0	1	0
0	0	1	0	0	0	1	1
0	1	1	0	0	1	0	0
0	1	1	1	0	1	0	1
0	1	0	1	0	1	1	0
0	1	0	0	0	1	1	1
1	1	0	0	1	0	0	0
1	1	0	1	1	0	0	1
1	1	1	1	1	0	1	0
1	1	1	0	1	0	1	1
1	0	1	0	1	1	0	0
1	0	1	1	1	1	0	1
1	0	0	1	1	1	1	0
1	0	0	0	1	1	1	1

K-Map for each binary output and get simplified expression

For B0

		G1 G0			
		00	01	11	10
G3	00	0	1	0	1
	01	1	0	1	0
G2	11	0	1	0	1
	10	1	0	1	0

$$B_0 = \bar{G}_3 \bar{G}_2 \bar{G}_1 G_0 + \bar{G}_3 \bar{G}_2 G_1 \bar{G}_0 + \bar{G}_3 G_2 \bar{G}_1 \bar{G}_0 + \bar{G}_3 G_2 G_1 G_0 + G_3 \bar{G}_2 \bar{G}_1 G_0 + G_3 \bar{G}_2 G_1 \bar{G}_0 + G_3 G_2 \bar{G}_1 \bar{G}_0 + G_3 G_2 G_1 G_0$$

$$B_0 = \bar{G}_3 \bar{G}_2 (G_1 \oplus G_0) + \bar{G}_3 G_2 (\bar{G}_1 \oplus \bar{G}_0) + G_3 \bar{G}_2 (G_1 \oplus G_0) + G_3 G_2 (\bar{G}_1 \oplus \bar{G}_0)$$

$$B_0 = (G_1 \oplus G_0) (\bar{G}_3 \bar{G}_2 + G_3 G_2) + (\bar{G}_1 \oplus \bar{G}_0) (\bar{G}_3 \bar{G}_2 + G_3 G_2)$$

$$B_0 = (G_1 \oplus G_0) (\bar{G}_3 \oplus G_2) + (\bar{G}_1 \oplus \bar{G}_0) (G_3 \oplus G_2)$$

$$B_0 = G_3 \oplus G_2 \oplus G_1 \oplus G_0$$

For B1

		G1 G0			
		00	01	11	10
G3	00	0	0	1	1
	01	1	1	0	0
G2	11	0	0	1	1
	10	1	1	0	0

$$B_1 = \bar{G}_3 \bar{G}_2 \bar{G}_1 + G_3 \bar{G}_2 \bar{G}_1 + \bar{G}_3 \bar{G}_2 G_1 + G_3 \bar{G}_2 G_1$$

$$B_1 = \bar{G}_1 (\bar{G}_3 \bar{G}_2 + G_3 \bar{G}_2) + G_1 (\bar{G}_3 \bar{G}_2 + G_3 \bar{G}_2)$$

$$B_1 = \bar{G}_1 (G_3 \oplus G_2) + G_1 (\bar{G}_3 \oplus \bar{G}_2)$$

$$B_1 = G_1 \oplus G_2 \oplus G_3$$

For B2

$G_3 \backslash G_2$	$G_1 G_0$	00	01	11	10
00	0	0	0	0	0
01	1	1	1	1	1
11	0	0	0	0	0
10	1	1	1	1	1

For B3

$G_3 \backslash G_2$	$G_1 G_0$	00	01	11	10
00	0	0	0	0	0
01	0	0	0	0	0
11	1	1	1	1	1
10	1	1	1	1	1

$$B_2 = G_3 \bar{G}_2 + \bar{G}_3 G_2$$

$$B_2 = G_3 \oplus G_2$$

$$B_3 = G_3$$

Gray Code

Bcd Code

